

Project 2: Exploratory Data Analysis (EDA)

Industrial Training Kit - DecodeLabs | Prepared by Tripti Mishra

Executive Summary

This report presents an exploratory analysis of 1200 order records from 2023-01-01 to 2025-06-30. The objective was to calculate descriptive statistics, identify trends and outliers, and summarize key business observations. Total revenue is INR 1,264,761.96, average order value is INR 1,053.97, and the highest revenue month is 2024-06.

KPI	Value
Total Records	1,200
Total Revenue	INR 1,264,761.96
Average Order Value	INR 1,053.97
Date Range	2023-01-01 to 2025-06-30
Top Product by Revenue	Chair
TotalPrice Outliers	8

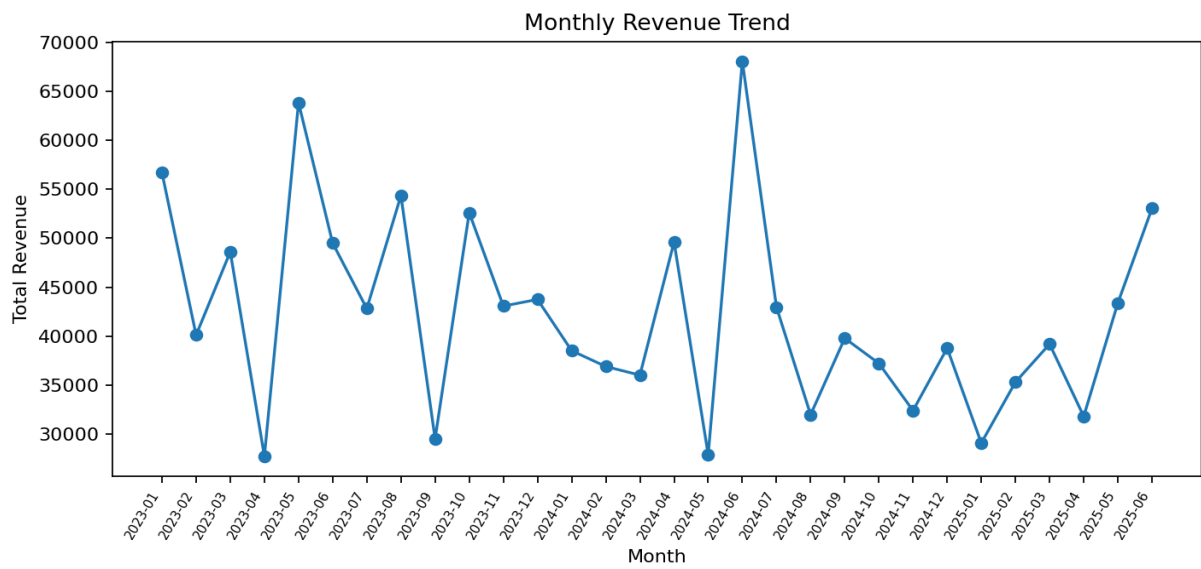
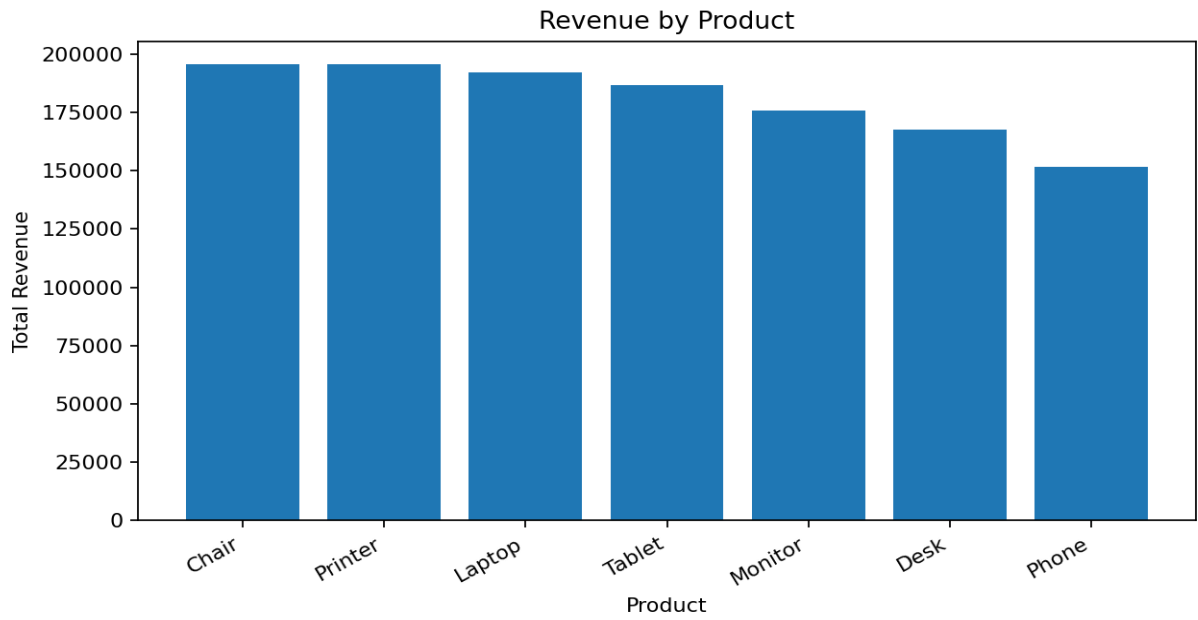
Methodology

- Converted date values into consistent yyyy-mm-dd format.
- Filled missing CouponCode values with NO_COUPON to preserve all transaction rows.
- Validated duplicate OrderIDs and TotalPrice = Quantity x UnitPrice.
- Calculated mean, median, min, max, quartiles, standard deviation and IQR outlier bounds.
- Created category-wise and month-wise summaries for business interpretation.

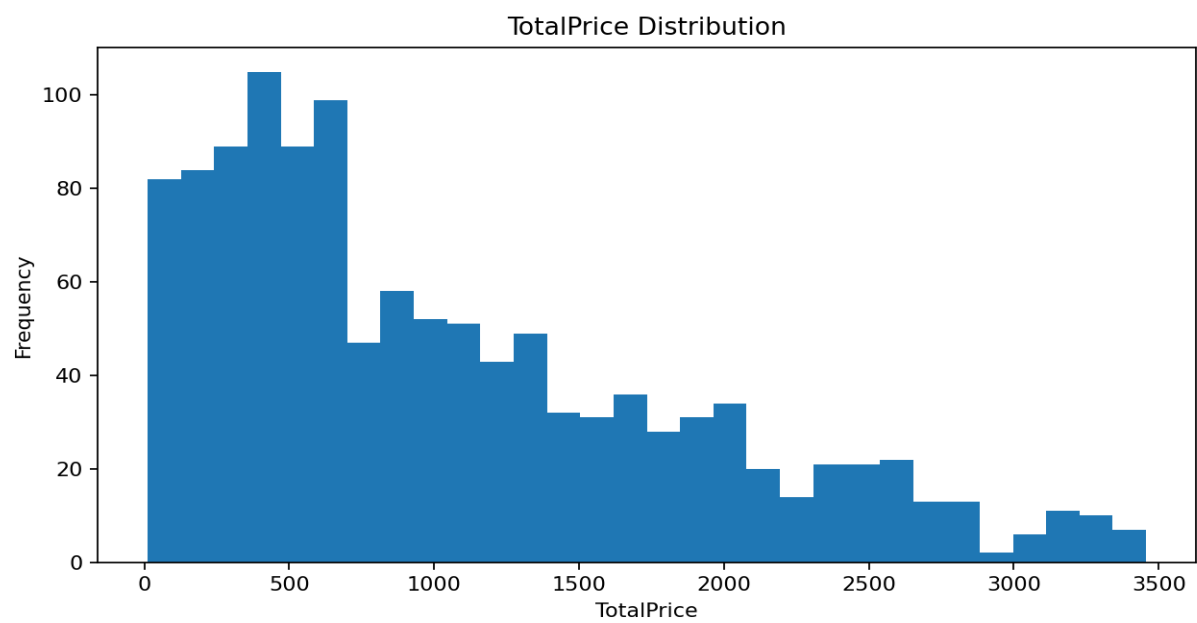
Descriptive Statistics

Column	Count	Mean	Median	Min	Max	Outliers
Quantity	1200	2.95	3.00	1.00	5.00	0
UnitPrice	1200	356.41	364.21	11.39	699.93	0
ItemsInCart	1200	5.49	5.00	1.00	10.00	0
TotalPrice	1200	1053.97	823.62	11.39	3456.40	8

Visual Analysis



Distribution and Outlier Analysis



Using the IQR method, TotalPrice has 8 high-value outlier records. These records were not removed because they appear to be valid high-value business transactions rather than data errors.

Category-Wise Findings

Rank	Product	Orders	Revenue	Quantity
1	Chair	178	INR 195,620.11	562
2	Printer	181	INR 195,612.61	542
3	Laptop	173	INR 192,126.56	535
4	Tablet	179	INR 186,568.95	497
5	Monitor	163	INR 175,651.41	480

Correlation Analysis

Variable 1	Variable 2	Correlation	Interpretation
Quantity	UnitPrice	0.015	Weak positive
Quantity	ItemsInCart	0.650	Moderate positive
Quantity	TotalPrice	0.615	Moderate positive
UnitPrice	ItemsInCart	0.001	Weak positive
UnitPrice	TotalPrice	0.717	Strong positive
ItemsInCart	TotalPrice	0.393	Weak positive

Key Observations

- Chair is the highest revenue product, followed closely by Printer.
- 2024-06 shows the highest monthly revenue.
- UnitPrice has the strongest relationship with TotalPrice, followed by Quantity.
- Missing CouponCode values represent customers who purchased without a coupon; this group should be tracked separately as NO_COUPON.
- Cancelled and returned orders have meaningful revenue values, so order status should be considered before final sales reporting.

Recommendations

- Create product-level campaigns around top revenue products.
- Investigate high-value TotalPrice outliers as potential premium buyers or bulk purchase behavior.
- Track NO_COUPON customers separately to understand discount-independent purchasing.
- Use monthly revenue trends to plan inventory and marketing around high-performing periods.
- Before dashboarding, filter status where required, especially for Delivered/Shipped versus Cancelled/Returned records.

Deliverables Included

1. Cleaned Excel workbook with analysis sheets, charts, validation summary and change log.
2. PDF report containing methodology, statistics, visuals, findings and recommendations.
3. README file explaining the submission package.